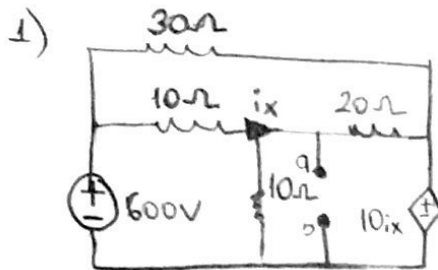
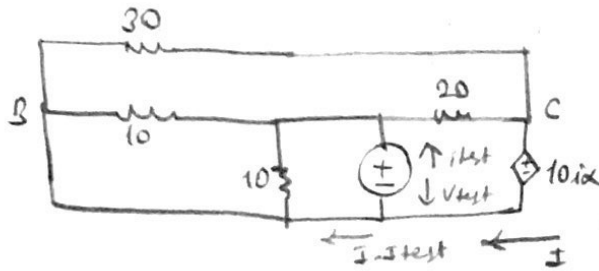


→ CEN-124 Circuit Theory Final Exam

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Find the thevenin equivalent seen from the terminals "a" and "b" of the following circuit



KCL at A

$$\frac{A-B}{10} + \frac{A}{10} + \frac{A-C}{20} = I_{test}$$

eq-1 $5A - 2B - C = 20I_{test}$

KCL at C

$$\frac{C-B}{30} + \frac{C-A}{20} = I \Rightarrow 5C - 2B - 3A = 60I$$

eq-2

KCL at B

$$\frac{B-C}{30} + \frac{B-A}{10} = I - I_{test} + \frac{A}{10}$$

$$4B - C - 3A = 30(I - I_{test}) + 3A$$

$$4B - C - 6A = 30(I - I_{test})$$

eq-3

dependent source voltage

$$V_C = \frac{B-A}{10} \times 10 = B-A=C$$

$$B = A+C$$

eq-1

$$5A - 2(A+C) - C = 20I_{test}$$

$$3A - 3C = 20I_{test}$$

eq-2

$$5C - 2A - 2C - 3A = 60I$$

$$3C - 5A = 60I$$

eq-2

eq-3

$$4A + 4C - C - 6A = 30I - 30I_{test}$$

$$3C - 2A = 30I - 30I_{test}$$

$$\Rightarrow 3C - 2A + 30I_{test} = 30I$$

$$\Rightarrow \text{eq-2} \Rightarrow 3C - 5A = 2(3C - 2A + 30I_{test})$$

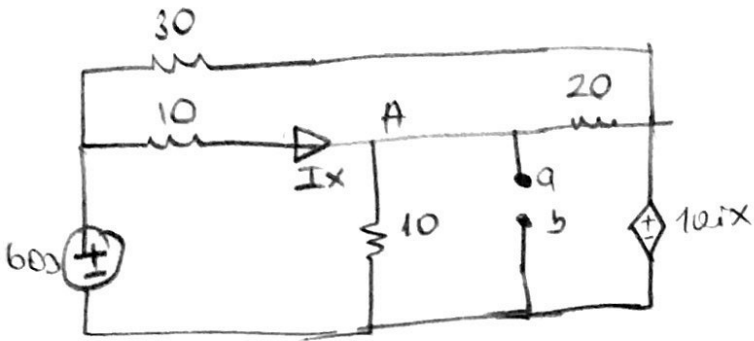
$$\text{eq-1} \quad 60I_{test} = -3C - A$$

$$3A + A = 20I_{test} - 60I_{test} \Rightarrow 4A = -40I_{test}$$

$$A = V_{test} \Rightarrow \frac{V_{test}}{V_{test}} = -10$$

$$R_{th} = 10\Omega$$

to find V_{th}



$$I_x = \frac{600 - A}{10}$$

$$10I_x = 600 - A \quad \text{eq 1}$$

KCL at A

$$\frac{A - 600}{10} + \frac{A}{10} + \frac{A - 10I_x}{20} = 0$$

$$5A - 1200 - 10I_x = 0 \quad \text{eq 2}$$

solve eq 1 and 2

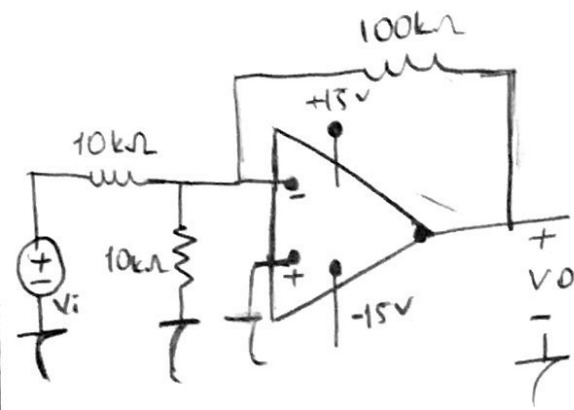
$$5A = 1200 + 10I_x \Rightarrow 1200 + 10I_x = 3000 - 50I_x$$

$$60I_x = 1800$$

$$I_x = 30 \text{ Amper}$$

2) (30) a) Find V_o in terms of V_i

10) b) Find the range of V_i that the op-amp operates in its linear region.

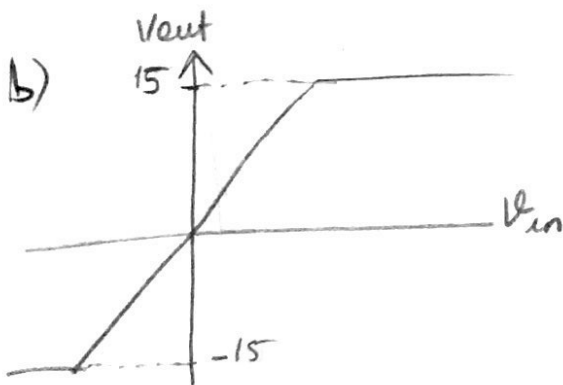


a) In linear region

$$V_A = V^- = V^+ = 0 \text{ volt}$$

$$\Rightarrow I_1 = I_2 = I_3 \text{ by KCL} \quad \frac{V_i - V_A}{10k} - \frac{V_A}{10k} = \frac{V_A - V_o}{100k}$$

$$\Rightarrow \frac{V_i}{10k} - 0 = \frac{-V_o}{100k} \Rightarrow V_o = -10V_{in}$$



$$\Rightarrow \text{When } V_o = 15V \\ V_{in} = -150V$$

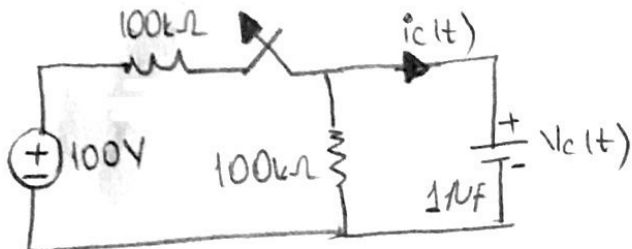
$$\text{When } V_o = -15V \\ V_{in} = +150V$$

$$\Rightarrow \text{range of } V_{in} = [-150, 150]$$

3) The switch has been closed for a long time before opening (disconnecting) at time $t=0$

(30) a) $V_c(t) = ?$ for $t > 0$

(10) a) $i_c(t) = ?$ for $t > 0$



When switch is closed $V_{c\text{ initial}} = V_{\text{source}} = -100\text{V}$

After switch is opened $V_{c\text{ final}} = 0\text{ volt}$

$$\Rightarrow a) V_c(t) = V_f + (V_i - V_{f\text{ final}}) \cdot e^{-t/\tau}$$

$$\text{where } \tau = RC = 100\text{k}\Omega \times 10^{-6}$$

$$= 100^{-1} = 0,1$$

$$\Rightarrow V_c(t) = 0 + (-100 - 0) \cdot e^{-10t}$$

$$V_c(t) = -100e^{-10t}$$

$$b) i_c(t) = C \frac{dV_c}{dt} \Rightarrow i_c(t) = 10^{-6} \times (-100) \times (-10) \cdot e^{-10t}$$

$$= -10^{-3} \cdot e^{-10t} \text{ Amper}$$